1. MATTRICES

1.1.1: Definition of a matrix

A matrix is an ordered rectangular array of numbers (or functions). Examples:

$\begin{bmatrix} 3 & -2 \\ 5 & -4 \end{bmatrix}$, $\begin{bmatrix} 6 & -1 & 3 \\ 1 & 5 & -1 \\ 4 & 3 & 1 \end{bmatrix}$, $\begin{bmatrix} 6 & -1 & 3 & 7 & -4 \\ 1 & 5 & -1 & -8 & 11 \end{bmatrix}$ and $\begin{bmatrix} a & b & c & d \\ e & f & g & h \end{bmatrix}$ are all matrices.

The numbers / letters enclosed inside the braces are referred to as the elements of the matrix.

The horizontal lines of elements are said to constitute rows of the matrix and the vertical lines of elements are said to constitute columns of the matrix.

1.1.2 Definition: Order of a Matrix

A matrix having m rows and n columns is called a matrix of order $m \times n$ or simply an $m \times n$ matrix (read as an "m by n matrix").

1.1.3: Denotation of matrices

We name matrices using uppercase letters (we may sometimes bold them). Elements which are letters are always written using lower case letters.

1.1.4: *Order of matrices examples and exercise*

i) The matrix $\mathbf{A} = \begin{bmatrix} -4 & 5 \\ -8 & 9 \end{bmatrix}$ has $\underline{\quad - - - \quad}$ rows and $\underline{\quad - - - \quad}$ collumns. Its a matrix of order $\underline{\quad - - - \quad}$, or an $\underline{\quad - - - \quad}$ matrix.

ii) The matrix $\mathbf{B} = \begin{bmatrix} 4 & 3 & 1 \\ 1 & 5 & -1 \\ 6 & -1 & 3 \end{bmatrix}$ has $\underline{\quad - - - \quad}$ rows and $\underline{\quad - - - \quad}$ collumns. Its a matrix of order $\underline{\quad - - - \quad}$, or an $\underline{\quad - - - \quad}$ matrix.

iii) The matrix $\mathbf{C} = \begin{bmatrix} a & b & c & d \\ e & f & g & h \end{bmatrix}$ has $\underline{\quad - - - \quad}$ rows and $\underline{\quad - - - \quad}$ collumns. Its a matrix of order $\underline{\quad - - - \quad}$, or an $\underline{\quad - - - \quad}$ matrix.

iv) The matrix $\mathbf{D} = \begin{bmatrix} -4 & 5 & \pi \end{bmatrix}$ has $\underline{\quad - - - \quad}$ rows and $\underline{\quad - - - \quad}$ collumns. Its a matrix of order $\underline{\quad - - - \quad}$, or an $\underline{\quad - - - \quad}$ matrix. Its referred to as a $\underline{\quad - - - \quad}$ matrix.

v) The matrix $\mathbf{E} = \begin{bmatrix} -6 \\ 5 \\ -8 \\ 9 \end{bmatrix}$ has $\underline{\quad - - - \quad}$ rows and $\underline{\quad - - - \quad}$ collumns. Its a matrix of order $\underline{\quad - - - \quad}$, or an $\underline{\quad - - - \quad}$ matrix. Its referred to as a $\underline{\quad - - - \quad}$ matrix.

vi) In general, an $m \times n$ matrix, denoted as $[A]_{ij}$, has the following rectangular array:

$$[A]_{ij} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} \cdots & a_{mn} \end{bmatrix},$$

where $i = 1,\ 2,\ 3,\ \cdots,\ n$ and $j = 1,\ 2,\ 3,\ \cdots,\ m$.

1.1.4: *Order of matrices exercise*

State the order of the following matrices:

i) $\mathbf{A} = \begin{bmatrix} 9 & 5 & 8 \\ -8 & -5 & 12 \end{bmatrix}$.

ii) $\mathbf{B} = \begin{bmatrix} 2 & 3 & -4 & -7 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$.

iii) $\mathbf{C} = \begin{bmatrix} 2 & 3 & -4 & -7 \\ 0 & -7 & -3 & 4 \\ 3 & -1 & -4 & 3 \\ 8 & 1 & -7 & -1 \\ 0 & -7 & -3 & -11 \\ 8 & 7 & -4 & 17 \end{bmatrix}$.

iv) $\mathbf{D} = \begin{bmatrix} -6 \\ 5 \\ -8 \\ 9 \\ 0 \\ -9 \\ -4 \end{bmatrix}$.

v) $\mathbf{E} = \begin{bmatrix} p & q & r & s & t & m & n & q \end{bmatrix}$.

vi) Write down a matrix of order: (a) 6×3 (a) 4×7

1.1.5: *Some Remarks*

The element a_{ij} is an element lying in the ith row and j^{th} column and is known as the $(i-j)^{th}$ element of matrix $\mathbf{A}$. The number of elements in an $m \times n$ matrix will be equal to mn.

1.1.6: *An Exercise*

a) Given the matrix $\mathbf{B} = \begin{bmatrix} -7 & 3 & 8 & 1 & -7 & -8 \\ 2 & 3 & -4 & -7 & -3 & 15 \\ -21 & 1 & 4 & 3 & -1 & -4 \\ -1 & 8 & 6 & -19 & 56 & -13 \end{bmatrix}$, identify each of the given elements:

b) Write a 3×5 matrix $\mathbf{B}$ whose elements are give (in no order at all!!!) as .....

1.2: *TYPES OF MATRICES / SPECIAL MATRICES*

I) A matrix is said to be a row matrix if it has only one row.

II) A matrix is said to be a column matrix if it has only one column.

III) A matrix in which the number of rows are equal to the number of columns, is said to be a square matrix. Thus, an $m \times n$ matrix is said to be a square matrix if $m = n$, and is known as a square matrix of order n.

IV) A square matrix $A = [\, A_{ij} \,]_{n \times n}$ is said to be a diagonal matrix if its all non - diagonal elements are zero, that is a matrix $A = [\, A_{ij} \,]_{n \times n}$ is said to be a diagonal matrix if $a_{ij} = 0$ whenever $i \neq j$.

V) A diagonal matrix B is said to be a scalar matrix if its diagonal elements are equal, that is, a square matrix $B = [\, B_{ij} \,]_{n\times n}$ is said to be a scalar matrix if $b_{ij} = 0$ whenever $i \neq j$ and $a_{ij} = k$ whenever $i = j$ for some constant k.

VI) A square matrix I_n in which elements in the diagonal are all 1 and rest are all zeroes is called an identity matrix. In other words the square matrix of order n I_n is said to be an identity matrix if $i_{jk} = 0$ whenever $j \neq k$ and $i_{jk} = 1$ whenever $j = k$.

VII) A matrix O_n is said to be zero matrix or null matrix of order n if all its elements are zeroes.

VIII) Two matrices $A = [\, A_{ij} \,]_{n\times n}$ and $B = [\, B_{ij} \,]_{n\times n}$ are said to be equal if

i) They are of the same order, and

ii) Each element of A is equal to the corresponding element of B, that is, $a_{ij} = b_{ij}$.

1.3: *OPERATIONS ON MATRICES*

We may add, subtract multiply by a scalar or find a product of two matrices as explained below.

1.3.1: *Addition and Subtraction of matrices*

Two matrices can be added (or subtracted)if they are of the same order. We add or subtract two matrices by adding or subtracting corresponding elements i.e

$$\mathbf{A} \pm \mathbf{B} = [A]_{ij} \pm [B]_{ij} = \begin{bmatrix} a_{11} & a_{12} & a_{13}\cdots & a_{1n} \\ a_{21} & a_{22} & a_{23}\cdots & a_{2n} \\ a_{31} & a_{32} & a_{33}\cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3}\cdots & a_{mn} \end{bmatrix} \pm \begin{bmatrix} b_{11} & b_{12} & b_{13}\cdots & b_{1n} \\ b_{21} & b_{22} & b_{23}\cdots & b_{2n} \\ b_{31} & b_{32} & b_{33}\cdots & b_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & b_{m3}\cdots & b_{mn} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} & a_{13} \pm b_{13}\cdots & a_{1n} \pm b_{1n} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} & a_{23} \pm b_{23}\cdots & a_{2n} \pm b_{2n} \\ a_{31} \pm b_{31} & a_{32} \pm b_{32} & a_{33} \pm b_{33}\cdots & a_{3n} \pm b_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} \pm b_{m1} & a_{m2} \pm b_{m2} & a_{m3} \pm b_{m3}\cdots & a_{mn} \pm b_{mn} \end{bmatrix}.$$

1.3.1: *Multiplication of Matrix by a Scalar*

Let $[A]_{ij}$ be an $m \times n$ and k is a scalar. Then kA is another matrix (called the scalar multiple of A) which is obtained by multiplying each element of A by a scalar k,

$$kA = k[A]_{ij} = \begin{bmatrix} ka_{11} & ka_{12} & ka_{13}\cdots & ka_{1n} \\ ka_{21} & ka_{22} & ka_{23}\cdots & ka_{2n} \\ ka_{31} & ka_{32} & ka_{33}\cdots & ka_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ ka_{m1} & ka_{m2} & ka_{m3}\cdots & ka_{mn} \end{bmatrix}.$$

1.3.2: *Addition, Subtraction and scalar multiplication of matrices example*

$$\begin{bmatrix} 3 & 8 & 1 & -7 \\ 3 & -4 & -7 & -3 \end{bmatrix} + \begin{bmatrix} 8 & -5 & 6 & 4 \\ 2 & -4 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3+8 & 8+(-5) & 1+1 & -7+4 \\ 3+2 & -4+(-4) & -7+11 & -3+12 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 3 & 2 & -3 \\ 5 & -8 & 4 & 9 \end{bmatrix}$$

$$4\begin{bmatrix}2 & -4 & 11 & 12\\ 8 & -5 & 6 & 4\end{bmatrix} - 2\begin{bmatrix}3 & -4 & -7 & -3\\ 5 & 8 & 1 & -7\end{bmatrix} = \begin{bmatrix}8 & -16 & 44 & 48\\ 32 & -20 & 24 & 16\end{bmatrix} - \begin{bmatrix}6 & -8 & -14 & -6\\ 10 & 16 & 2 & -14\end{bmatrix}$$

$$= \begin{bmatrix}8-6 & -16-(-8) & 44-(-14) & 48-(-6)\\ 32-10 & -20-16 & 24-2 & 16-(-14)\end{bmatrix}$$

$$= \begin{bmatrix}2 & -8 & 58 & 54\\ 22 & -36 & 22 & 30\end{bmatrix}$$

1.3.3: ***Addition, Subtraction and scalar multiplication of matrices exercise***

a) $$2\begin{bmatrix}4 & 8 & 1\\ 8 & 1 & -7\\ 3 & 5 & -6\\ 9 & 13 & -7\end{bmatrix} + 3\begin{bmatrix}-2 & 5 & 0\\ 6 & 10 & -4\\ 11 & -1 & -8\\ 2 & -15 & 5\end{bmatrix}$$

b) $$5\begin{bmatrix}-7 & 13 & 4 & 3\\ 3 & 8 & 1 & 17\\ -6 & 3 & 8 & 9\end{bmatrix} - 3\begin{bmatrix}-5 & 4 & 10 & 16\\ 7 & -11 & 1 & 6\\ 3 & 4 & 8 & -7\end{bmatrix}$$

c) Solve the matrix equation: $2\left(\begin{bmatrix}2x & -3\\ 5 & -y\end{bmatrix} + \begin{bmatrix}-1 & 4\\ 3 & 5\end{bmatrix}\right) = \begin{bmatrix}-18 & 2\\ 16 & 4\end{bmatrix}$

1.1.4: ***Square matrices examples Order of matrices exercise***

1) Find the Reduced Row Echelon Form of matrix $\mathbf{B}$ given $\mathbf{B} = \begin{bmatrix}2 & 3 & -4 & -7 & -3\\ 1 & 4 & 3 & -1 & -4\\ 1 & 3 & 1 & -2 & -3\\ 3 & 8 & 1 & -7 & -8\end{bmatrix}$ **(5 marks)**

2) Write down a 3 x 3 matrix whose trace is equal to -7. **(1 mark)**

3) Use matrix inversion to solve the linear system below. **(4 marks)**

$$2x - 4y = -14$$

$$x + 5y = 7$$

4) Use the cofactors of $\mathbf{B}$ to find $\mathbf{B}^{-1}$ given that $\mathbf{B} = \begin{bmatrix} 6 & -1 & 3 \\ 1 & 5 & -1 \\ 4 & 3 & 1 \end{bmatrix}$. **(6 marks)**

5) Find the value of r and s for which the matrix $\mathbf{B} = \begin{bmatrix} 0 & 7 & 3r-s \\ s & 0 & -5 \\ -11 & 5 & 0 \end{bmatrix}$ is skew - symmetric. **(3 marks)**

6) Use row operations to evaluate the inverse of $\mathbf{B} = \begin{bmatrix} 3 & -2 \\ 5 & -4 \end{bmatrix}$. **(4 marks)**

7) Use Cramer's rule to find the values of x, y and z in the linear system below. **(6 marks)**

$$x + 2y + 3z = 5$$
$$4x + 5y + 6z = 14$$
$$5x + 7y + 8z = 17$$

8) Evaluate the determinant of $\mathbf{E}$ by row reduction given $\mathbf{D} = \begin{bmatrix} 5 & -2 & 8 \\ 1 & 2 & -3 \\ 4 & -1 & 6 \end{bmatrix}$. **(4 marks)**